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Innovative Well Rejuvenation: UAE's First E-Line Electromechanical Intervention and ADNOC Onshore's Maiden Fiber Optic Coiled Tubing ICD Manipulation Unlocks Production Potential

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Abstract

This paper shares key lessons from the UAE's first e-line electromechanical and ADNOC Onshore's first fiber-optic equipped coiled tubing (FOE-CT) Inflow control device (ICD) shifting operations in 2024. It shows how real-time telemetry and distributed temperature surveying (DTS) enabled precise ICD assessment and manipulation, overcoming limitations of conventional CT. The work highlights production gains, tool design advances, and sustainability impact. It introduces a strategic approach to revitalize legacy wells affected by high water production, setting the stage for future optimization using advanced diagnostics and interventions.

The paper outlines a data-driven methodology applied in the UAE's first e-line electromechanical and FOE-CT ICD shifting operations in 2024. It includes technical evaluation of hydraulic manipulation tools, real-time telemetry, and DTS compared to conventional CT. Candidate wells were selected through production log analysis and backlog review. Procedures involved detailed pre-job planning, tool deployment with real-time instrumentation, hydraulic ICD manipulation, and post-job status verification using logging and DTS. Production uplift was measured to assess intervention effectiveness. This integrated approach demonstrates operational feasibility and sets a scalable framework for revitalizing legacy wells with inflow control challenges and high-water production.

The 2024 operations involving UAE's first e-line electromechanical shifting and our organization's inaugural FOE-CT ICD shifting job, delivered transformative outcomes, and boosting average production per well. The e-line instrumented electromechanical shifting tool's precise control and millimeter accuracy as well as FOE-CT's real-time telemetry and hydraulic piston control ensured precise shifting, eliminating excessive surface overpull that plagued conventional CT's ability of ICD shifting since 2009. Real-time displacement measurement and force signatures confirmed successful ICD actuation, corroborated by immediate pre- and post-manipulation logging, achieving 100% verification accuracy via downhole sensors. Observations highlighted the technologies' ability to overcome unquantified downhole force

limitations, ensuring reliable, repeatable operations. Benefits included precise manipulation, enhanced operational efficiency, and validated ICD status, unlocking previously inaccessible production potential. Conclusions affirm the scalability of these breakthroughs for over 15 years of unmanipulated ICDs leading to unoptimized production. This paradigm-shifting framework, leveraging real-time data and precision engineering, sets a global benchmark for ICD optimization, promising substantial production gains in legacy assets worldwide.

This paper introduces a validated workflow for ICD manipulation using the UAE's first e-line electromechanical and FOE-CT ICD shifting operations, achieving millimeter-scale precision and real-time verification. It demonstrates how DTS enables accurate ICD position assessment and immediate fluid profiling. Unlike conventional CT, these methods avoid blind overpull and provide actionable data. Case studies and financial analysis confirm cost-effectiveness and scalability, offering a breakthrough model for revitalizing legacy ICD wells and enhancing global completion optimization strategies.

INTRODUCTION

The integration of ICDs in horizontal well completions offers a strategic approach to managing water production and enhancing oil recovery, especially in mature reservoirs. According to [Murtaza and Madan \(2024\)](#), sophisticated completions with ICDs utilize open-hole packers to segment the horizontal liner into compartments based on reservoir characteristics identified through open-hole logs. This compartmentalization enables targeted fluid control, critical for mitigating water production. However, achieving success in water shut-off (WSO) operations requires a comprehensive evaluation of the completion and production data before deploying CT or e-line tractor conveyed shifting tools to manipulate ICDs. The objective is to optimize oil production while minimizing water influx, recognizing that water, either from aquifers or injection operations, is the primary driver that brings oil from the formation into the wellbore. Over time, excessive water production can make oil extraction economically unattractive or halt production entirely due to a high water cut. Active ICDs, which can be opened or closed to control fluid flow from specific zones, provide operators with flexibility to adjust the flow profile dynamically.

The manipulation of ICDs plays a pivotal role in managing uneven production profiles and delaying the onset of water breakthroughs. As [Sharma et al. \(2020\)](#) highlight, horizontal wells are prone to imbalanced production along their extended intervals due to reservoir heterogeneities, such as fractures, permeable streaks, or voids, especially in carbonate formations. Active ICDs, manipulated using CT or electric-powered tractors equipped with modular anchors and shifting tools, help equalize flow and improve reservoir drainage efficiency. Operators use production logs to pinpoint water-producing zones before adjusting ICDs.

PROBLEM STATEMENT

For over 15 years, the manipulation of Inflow Control Devices (ICDs) in legacy horizontal wells across our organization and its sister organization in the UAE has been hindered by significant operational challenges using conventional coiled tubing (CT). Multiple attempts to shift ICDs in these horizontal wells, even by vendors who installed them, consistently failed due to the inherent limitations of conventional CT methods. The primary issue was the reliance on blind surface overpull, where unquantified and potentially excessive surface-applied forces were used without real-time feedback on downhole conditions. This lack of precision and visibility into downhole forces, compounded by the complex trajectories of horizontal wells, led to unreliable ICD actuation, resulting in unmanipulated ICDs, unoptimized production, and high-water production. These failures prompted exploration of alternative completion types, as the industry struggled to address inflow control challenges in horizontal wells effectively. The absence of a reliable, data-driven method for ICD manipulation in these wells limited production potential, increased operational costs, and

posed sustainability challenges due to excessive water handling, underscoring the need for an innovative approach to revitalize legacy horizontal wells.

Understanding Root Cause: Limitations of Conventional Coiled Tubing in Legacy ICD Wells

The use of conventional coiled tubing (CT) for manipulating Inflow Control Devices (ICDs) in legacy horizontal wells has been hindered by a range of technical and operational shortcomings, particularly in the complex reservoir environments of the UAE. These limitations have consistently prevented effective ICD shifting, resulting in unoptimized production and persistent high-water production.

Key shortcomings include:

1. **Blind Overpull:** Conventional CT relies on applying surface forces without real-time measurement of downhole forces, leading to unpredictable force transmission, especially in the extended reach and tortuous trajectories of horizontal wells.
2. **Lack of Real-Time Telemetry:** The absence of downhole sensors or telemetry systems means operators cannot monitor tool performance, ICD engagement, or actuation status during operations, increasing the risk of failure.
3. **Imprecise Tool Control:** Conventional CT lacks the fine, millimeter-scale control required for precise ICD manipulation, often causing tool misalignment or insufficient force application.
4. **Inability to Verify ICD Status:** Without real-time diagnostics, such as Distributed Temperature Sensing (DTS) or downhole sensors, there is no immediate confirmation of successful ICD shifting, requiring costly post-operation logging.
5. **Force Transmission Inefficiencies:** The flexible nature of CT and frictional losses in horizontal wells reduce the effective force delivered to the ICD, particularly in deep horizontal wells, often resulting in incomplete or failed manipulations.
6. **Risk of Tool Sticking:** Complex well geometries increase the likelihood of CT tools becoming stuck, leading to operational delays and potential well damage.

These shortcomings have led to repeated failures across our industry, prompting exploration of alternative completion types. A representative example from our operations involved a horizontal well where a vendor-supplied ICD was identified via Production Logging Tool (PLT) analysis as a target for closure to reduce water influx. Despite deploying conventional CT with the same vendor's specified shifting tool and applying significant surface overpull, no indication of ICD movement was observed. Post-intervention production profiles remained unchanged, and a subsequent PLT confirmed the ICD had not shifted, underscoring the ineffectiveness of the approach. Similar failures have been documented across multiple wells in our sister organization, highlighting the systemic limitations of conventional CT. These challenges have resulted in unmanipulated ICDs, sustained water production, increased operational costs, and reduced hydrocarbon recovery, necessitating a transformative solution to rejuvenate legacy horizontal wells.

Novel Solution

To address suboptimal performance, innovation was required in the ICD shifting process. As [Sharma et al. \(2020\)](#) emphasize, real-time monitoring significantly enhances ICD manipulation, with tool engagement and load changes providing clear confirmation of operational effectiveness. While post-intervention production testing and surface flow measurements validate reductions in water cut, a detailed nodal analysis is essential to determine the optimal number of ICD closures. Excessive closures can compromise overall hydrocarbon's productivity. By isolating high-permeability zones and balancing inflow across the wellbore, ICDs play a critical role in extending well life, reducing operational costs, and enhancing oil

recovery, highlighting their importance in modern reservoir management. Accordingly, the engineering team reviewed existing resources to develop an improved, engineered workflow.

To overcome the persistent challenges of Inflow Control Device (ICD) manipulation in horizontal wells, our organization planned a groundbreaking workflow leveraging an electromechanical ICD shifting tool, designed to deliver unprecedented precision and address the shortcomings of conventional coiled tubing (CT), such as blind overpull, lack of real-time feedback, and unreliable force transmission in complex well trajectories. The planned approach centered on evaluating the e-line deployed electromechanical, instrumented shifting tool for its ability to provide millimeter-scale accuracy and fine control, with expectations of enabling precise ICD engagement and actuation. The tool's design incorporated advanced hydraulic manipulation capabilities and compatibility with real-time instrumentation, aiming to ensure reliable operations in the extended reach and tortuous geometries of horizontal wells. For the CT a hydraulic actuated shifting tool combined with the head tension sensor above it as part of the downhole sensor package. These novel solutions were expected to unlock production potential, reduce water cut, and establish a scalable framework for revitalizing legacy wells in the UAE and beyond.

Central to the planned workflow was the use of two distinct conveyance methods to ensure robust real-time communication with the shifting tools, each tailored to specific well conditions and applied to separate horizontal wells: (1) electric line (e-line) with a tractor and (2) Fiber Optic Enabled Coiled Tubing (FOE-CT).

For the e-line method, the plan involved deploying the tool via an e-line system paired with a downhole tractor to navigate complex well trajectories. The e-line provided continuous power and real-time telemetry, enabling precise monitoring of tool performance, downhole forces, and ICD actuation status. The tractor was anticipated to ensure accurate tool positioning, with pre-job simulations planned to optimize tractor settings and e-line tension. Real-time data acquisition for shifting displacement and force signatures was expected to confirm successful ICD manipulation, with additional post-job verification of well performance using Production Logging Tools (PLT) anticipated to achieve 100% accuracy in assessing ICD status and fluid profiling.

For the FOE-CT method, the plan involved integrating fiber optic cables within the coiled tubing to enable continuous monitoring of downhole conditions, including temperature and pressure, via Distributed Temperature Sensing (DTS). This approach was expected to deliver actionable data on tool engagement and ICD actuation, eliminating the uncertainties of conventional CT. The FOE-CT system was planned to provide precise manipulation of the shifting tool with detailed pre-job planning, including well trajectory analysis and tool configuration, to ensure operational reliability. Post-job logging and DTS were expected to verify ICD shifting and quantify production uplift, establishing a repeatable process. The planned workflow included a data-driven well selection process, leveraging production log analysis and backlog review to identify horizontal wells with high water production and unmanipulated ICDs. The hydraulic actuated shifting tool with the downhole sensor package and telemetry was evaluated against conventional CT methods, with expectations of superior precision and real-time capabilities. Operational procedures were designed to include pre-job simulations, tool deployment with real-time instrumentation, hydraulic ICD manipulation, and post-job verification using logging and DTS. The anticipated production uplift was expected to serve as a key metric for intervention success. By planning to apply e-line with tractor in one set of wells and FOE-CT in another, the approach aimed to demonstrate operational flexibility and scalability, addressing diverse challenges in legacy horizontal wells. The expected outcomes included precise ICD manipulation, enhanced operational efficiency, reduced water production, and a validated framework for global ICD optimization. This innovative solution was poised to set a new industry benchmark, offering a cost-effective, sustainable, and data-driven strategy to rejuvenate legacy wells and unlock previously inaccessible production potential.

Tool Configuration

E-Line Shifting-Specific Components

- Custom Shifting Key for Specific ICD Profiles: A specialized shifting key was developed to latch onto unique ICD profiles, ensuring precise engagement. The key was designed to extend radially with controlled force, navigating well restrictions and enabling reliable manipulation of targeted ICDs in horizontal wells.
- Intelligent, Instrumented Shifting Tool: An array of integrated sensors allowed for real-time monitoring and control of downhole parameters, such as shifting force, linear displacement, and shifting key pressure, providing instant feedback on tool performance and ICD actuation. It was expected to eliminate the uncertainties of conventional CT's blind operations, ensuring high-accuracy verification.
- High Force Linear Actuator: This module delivers precisely controlled high axial force for smooth ICD actuation, overcoming force transmission inefficiencies in horizontal wells. The approach of generating this force downhole, unlike conventional CT's reliance on mechanical overpull from surface, ensures that the required force can be transmitted to the ICD regardless of depth. The bi-directional capability enables multiple shifts in a single run with precision and versatility across multizone completions.
- Anchor Module: An anchor module locks the tool in position with high force across multiple arms, ensuring stable positioning during manipulation without stressing the tubing, critical for reliable operations in complex well trajectories. The inbuilt diameter measurement confirms the correct position of the anchor and that no tubing deformation occurs.
- Smart Shifting Tool: Paired with the shifting key and linear actuator, this module was planned to execute precise ICD movements, integrating with the downhole instrumentation for real-time displacement and force measurements to ensure reliable actuation.

E-Line Conveyance

- Slim Tractor: A slim, intelligent tractor drive section was planned to navigate the extended reach and tortuous paths of horizontal wells, ensuring precise tool positioning. Pre-job simulations were expected to optimize tractor settings and e-line tension, with continuous power and telemetry anticipated to confirm ICD status via post-job Production Logging Tool (PLT) analysis.



E-Line Smart Shifting - Downhole Sensors

The instrumented, electromechanical, high expansion e-line shifting tool development and features are detailed by [Mauchien et al. \(2019\)](#). The tool modules provide the following sensors.

- Wellbore temperature.
- Wellbore Pressure.
- Casing Collar Locator (CCL).
- Gamma Ray.

- Anchor force and diameter.
- Linear actuator force and displacement.
- Shifting tool internal hydraulic pressure.

The ability to verify and adjust the downhole shifting operation in real-time with multiple, independent sensors ensures that the engineer makes the correct decisions for the greatest chance of success. The data from the sensors are used to verify successful shifting with a high degree of confidence. During the operation, the tool telemetry transmits all data to surface via the eLine cable continuously. The observed sleeve linear displacement can be compared to the technical schematic of the target ICD to verify complete shifting. The internal shifting tool hydraulic pressure signature can confirm when the sleeve was engaged, when shifting was completed and when the shifting tool disengaged. Each module can either be simultaneously or individually controlled as required at each step of the operation. The array of independent downhole sensors allows for the removal of any doubt as to the final position of the ICD sleeve after the shifting operation with the additional ability for real-time, remotely monitored confirmation.



Fiber-Optic Equipped Coiled Tubing (FOE-CT)

The first improvement involved transitioning from conventional CT to a FOE-CT string. FOE-CT enables real-time downhole data acquisition, as the integrated fibers function as sensors for DTS along the wellbore.

- FOE-CT Deployment:
 - Fiber Optic Telemetry System: Fiber optic cables integrated within the coiled tubing were planned to enable continuous monitoring of downhole conditions, including temperature and pressure, via Distributed Temperature Sensing (DTS). This system was expected to deliver actionable data on tool engagement and ICD actuation, with post-job DTS and logging anticipated to verify production uplift.

Coiled Tubing Downhole Sensors

Downhole sensors, as described by [Anopov et al. \(2017\)](#) and [Alshehhi et al. \(2025\)](#), are attached to the end of the CT and include the following measurements:

- Wellbore Temperature.
- Pressure inside the CT.
- Pressure outside the CT.
- Casing Collar Locator (CCL).
- Gamma Ray.
- Tension, compression, and torsional loads.

The real-time readings provided by these sensors deliver critical insights into downhole dynamics during job execution. This data is essential for evaluating performance and ensuring operational efficiency, as will be demonstrated in the case study.

Distributed Temperature Surveying (DTS)

DTS provides a continuous temperature log when the CT is run through the tubing and along the horizontal lateral section of the wellbore. Before ICD manipulation, DTS assesses the initial completion condition, offering valuable data for evaluating water injection profiles both pre- and post-shifting of the ICDs. DTS also detects crossflows and other relevant wellbore events.

According to [Gurses and Kozin \(2018\)](#), the primary inputs for DTS analysis include:

- Well deviation survey.
- Completion schematic.
- Reservoir properties.
- Pressure/temperature gradients.
- DTS recorded data.
- Gamma ray and CCL logs improve the depth correlation between the wellbore completion and CT position ([Alshehhi, 2025](#)),

Active ICDs

[Fig. 1](#) illustrates the specific model of active ICD used in the case study. Those ICDs are installed in the lower completion and later they are manipulated to achieve the shut-off of unwanted fluids from the reservoir.

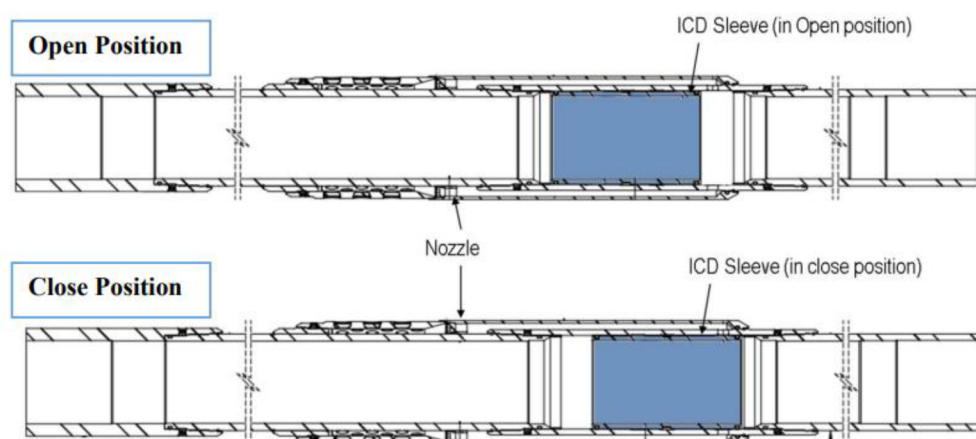


Figure 1—ICD illustration showing on top the ICD sleeve in open position, and on bottom the ICD sleeve in close position.

Theory of Operation.

- RIH shifting tool (HHST) with CT above the ICD sleeve.
- Begin extending HHST keys by pumping fluid across the tool generating a differential pressure of 1,000 to 1,500 psi.
- Push down the BHA through the valve to shift into closed position. Watch the weight indicator for signature of shifting 3,000 lbf is the "surface recorded value" of the required compressive force to close/open the ICD valve. [Fig. 1](#) displays the open/close positions of the ICD, and the blue area corresponds to the sleeve.

Coiled Tubing Shifting Tool Model HH

The shifting tool (HHST) utilized in this project offers several notable features ([Fig. 2](#)):

- Slim Design: Allows conveyance through multiple restrictions in the upper completion.

- **Unlimited Actuation:** Enables repeated opening and closing of ICD sleeves.
- **Versatile Deployment:** Can be deployed using either CT or jointed pipe.
- **Self-Centralizing Keys:** Geometry ensures automatic centralization when pumping is activated.
- **Hydraulic Actuation:** The valve operates hydraulically, enabling maximum overpull or compression forces without unintended release. Controlled shut off of the pumping unit ensures intentional tool disengagement.

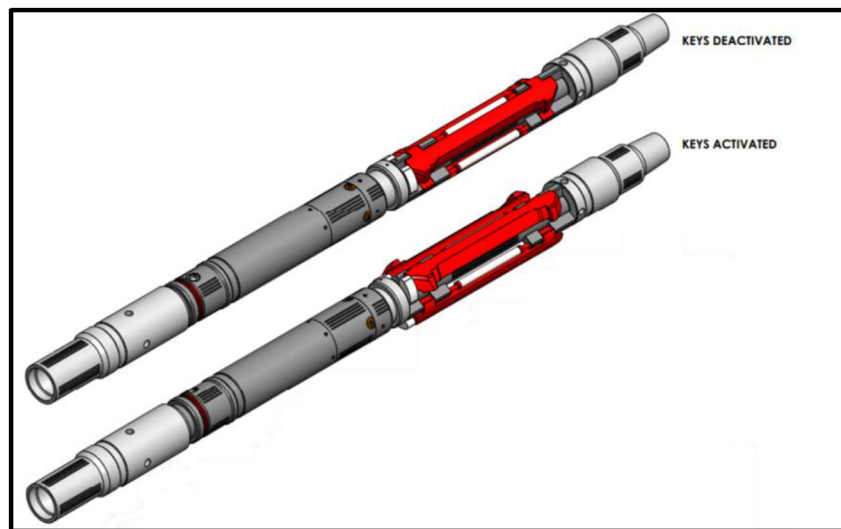


Figure 2—Slimhole shifting tool illustration showing on top the tool with the deactivated keys and in bottom the tool with activated keys.

These features significantly enhance operational flexibility and reliability during ICD interventions.

Operational Success

Well A – Intelligent ICD Shifting Operation with e-Line

In Well A, the intervention targeted three ICDs for mechanical shifting using a slim, sour service, instrumented intervention toolstring deployed via high-strength, torque balanced e-Line. The toolstring included an intelligent tractor conveyance system with independent opening and closing ability of each drive section, with a proven track record for deployment in such wells (Varghese et al. 2023). The e-Line tractor provided the conveyance force to deliver the instrumented, universal shifting tool to the target ICD depths. The shifting tool is composed of the following key modules, hydraulic linear actuator, anchor module, and smart shifting tool. The operation was conducted in a high deviation, ERD well with a completion featuring ICDs and swell packers that required tractor arm manipulation to successfully navigate.

Operational Steps

- **Tool Deployment:** The toolstring was deployed to the target ICD depth using intelligent tractor conveyance.
- **Depth Correlation:** CCL and gamma ray sensors were used to confirm tool position close to the target ICD.
- **Anchor Setting:** Anchors were engaged to fix the tool in position, confirming the ID measurement for correct location.

- Shifting Execution: The hydraulic linear actuator applied force to shift the ICD sleeves.
- Tool Instrumentation: Real-time linear displacement and force data were monitored to confirm actuation.

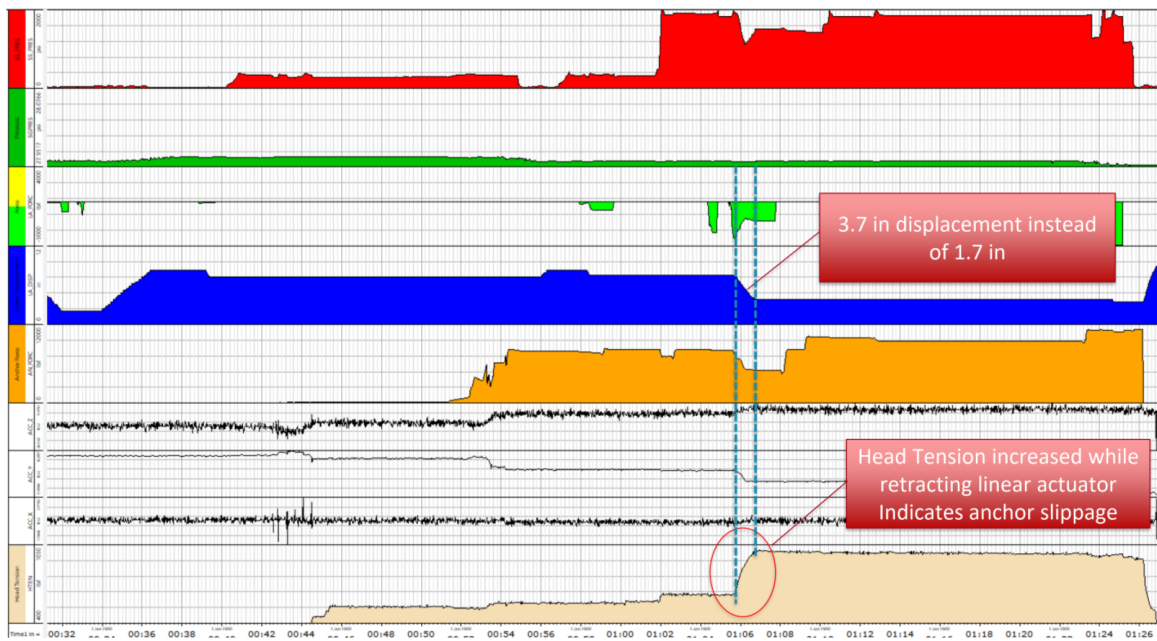
The shifting tool was conveyed to the target depth just below the ICD assembly, starting with the deepest target. The universal shifting tool was activated in a low-pressure mode to seek and locate the shifting profile. In this mode, the eLine BHA was moved upwards using the wireline winch to engage the shifting key on the sleeve profile. Once successfully engaged, the downhole anchor was set to lock the tools in position. The integrated anchor ID and force measurements confirm that the anchor is located in the expected tubing joint and that no slippage will occur during ICD manipulation. Once locked in place, the internal hydraulic pressure of the shifting tool is increased to ensure it will not disengage, then shifting force is applied via the linear actuator to move the sleeve to the desired position. While moving the sleeve to the closed position, the sensor array integrated into the tool is monitored for confirmation of successful shifting. The expected travel distance of the sleeve was known to be 1.6" based on the ICD technical schematics and this same distance should be observed by the linear displacement sensor. The shifting force was expected to be around 3,000lbf according to the manufacturer specification. The combination of internal shifting tool pressure, consistent anchor force and linear displacement matching the expected 1.6" act as multiple verification points to remove any doubt of the success of the shifting operation.

Execution Details

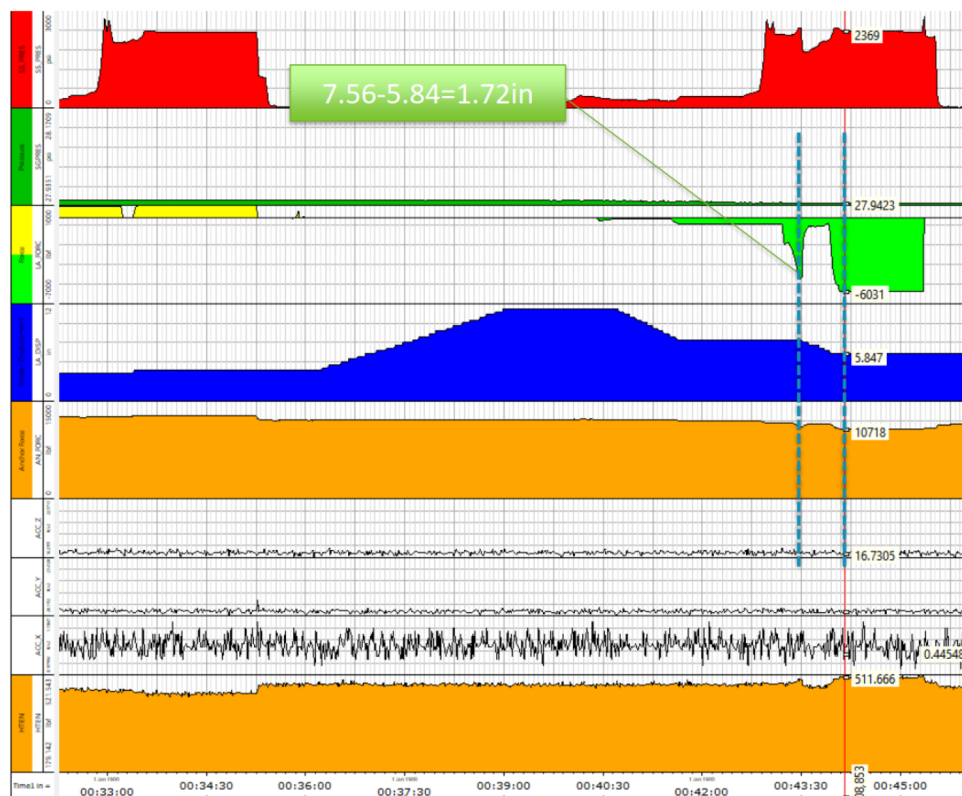
- ICD #8 (First Attempt): Anchor force dropped partially during shifting and some increase in the logging head tension sensor indicated slippage. Observed displacement was 3.7 inches vs expected ~1.7 in, leading to a second attempt.
- ICD #8 (Second Attempt): Set with increased anchor force, achieved precise 1.72 in displacement with ~6,000 lbs force, confirming successful shift.
- ICD #7: Shifted with 1.72 in displacement and consistent force profile.
- ICD #6: Shifted with 1.73 in displacement, confirming mechanical actuation.

During shifting of the initial target ICD, the anchor force was set to 7,000lbf and the linear actuator was activated to close the sleeve. A force limit of 5,000lbf was set on the linear actuator, once this limit was reached the sleeve should have fully moved. However, a displacement measurement of 3.7" was observed rather than the expected 1.7". Verifying multiple sensors allowed for real-time understanding of what had taken place. The anchor force dropped during the initial sleeve movement at around 5,000lbs actuator force. Simultaneously, an increase was observed at the logging head tension sensor. Both were indicators that the anchor had partially slipped during the sleeve manipulation as seen in the Figure below. It was still expected that the sleeve had fully closed, but due to this mismatch of displacement distance the sleeve shifting was repeated.

The shifting tool was re-positioned to push the sleeve down to re-open it, with an increased anchoring force of 12,000lbf. The shifting distance observed during re-opening was 1.7" as expected and there was no change in either anchor force or logging head tension sensor. This confirmed that the anchor had remained locked in place and the sleeve was completely open. Finally, the sleeve was re-closed using the same anchor force of 12,000lbs and once more the correct linear displacement of exactly 1.6" was observed. This additional manipulation and verification removed any doubt as to the status of this first ICD. The same parameters were used for the subsequent ICDs, as seen in the Figure below, and all 3 were successfully closed in a 6-hour operation.



Timeframe: The entire operation, including deployment, shifting, and verification, took approximately 6 hours.



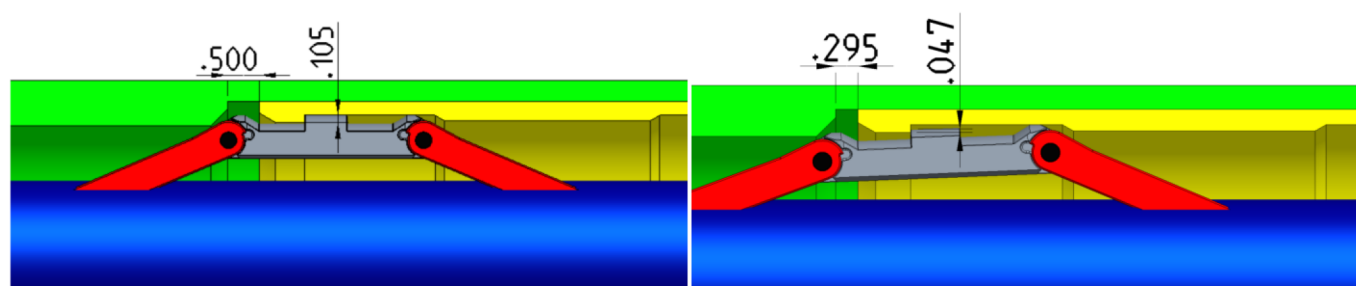
Post-Job Analysis

- Displacement accuracy within 0.1 inch confirmed precise actuation.

- Multiple, independent correlated measurements including internal shifting tool pressure, linear actuator force, liner actuator displacement, anchor force, logging head tension validated success closure.
- Force-displacement graphs showed clear mechanical engagement signatures.

Well B – Multi-Zone ICD Shifting with e-Line

In Well B, the intervention objective was to verify the current position and shift two ICDs from a total of eight distributed across a multi-zone completion from ~10,400 ft to ~14,150 ft MD. The well had an extended horizontal section with high inclination up to 89.4°, requiring the same slim, intelligent tractor conveyance and high-strength, torque balanced eLine cable as used in Well A. Once more, the presence of ID changes in the lower completion would require tractor arm manipulation to pass and successfully reach the target depth. An additional challenge for the ICD present in Well B was that it was not possible to perform a pre-job System Integration Test (SIT) as is normal practice due to the lack of availability of the corresponding ICD assembly. As such, a comprehensive 3D simulation was completed to ensure that the custom shifting key design would reliably shift the ICD sleeves. The entire ICD assembly was modelled in 3D and a simulation of the shifting operation was completed to ensure both that the engagement area of the key and profile was sufficient, also verifying that the key would not unintentionally latch on any part of the assembly. The Figure below shows some snapshots from this analysis to determine the behavior of the key during shifting and at the end of sleeve travel.



Operational Steps

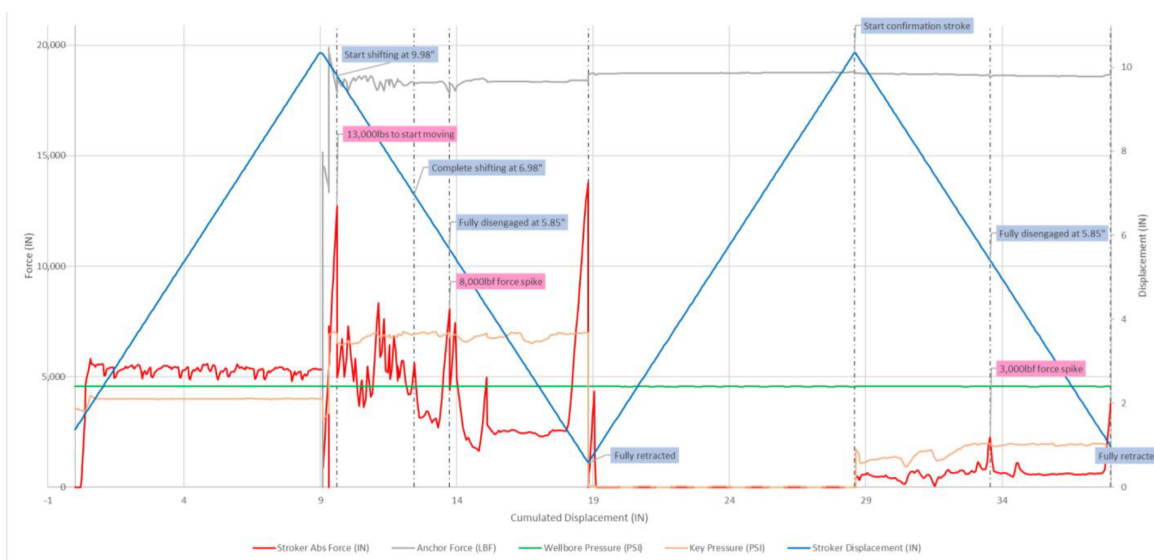
1. Tool Deployment: The toolstring was deployed to the target ICD depth using intelligent tractor conveyance.
2. Depth Correlation: CCL and gamma ray sensors were used to confirm tool position close to the target ICD.
3. Anchor Setting: Anchors were engaged to fix the tool in position, confirming the ID measurement for correct location
4. Shifting Execution: The hydraulic linear actuator applied force to shift the ICD sleeves.
5. Tool Instrumentation: Real-time linear displacement and force data were monitored to confirm actuation.

The shifting tool was again conveyed to the target depth just below the ICD assembly, starting with the deepest target. Following the same process as Well A, the universal shifting tool was activated in a low-pressure mode to seek and locate the shifting profile. In this mode, the eLine BHA was moved upwards using the wireline winch to engage the shifting key on the sleeve profile before the anchor was engaged to allow application of high force shifting via the linear actuator. For this sleeve, the expected travel distance was 3.0" with the same 3,000lbf, so the same parameters would be used based on the previous well experience using validation of multiple sensors to confirm successful shifting.

Execution Details

- All planned ICDs were successfully shifted.
- Displacement values precisely matched, reading 3.10 inches
- Shifting forces higher than expected with up to 13,000lbf to move sleeve
- Displacement and force measurements confirmed engagement and complete shifting
- Each shift was confirmed via real-time data analysis and post-job QC.

Due to the expected shifting force of around 3,000lbf, the anchor force was set to 12,000lbf and the linear actuator was activated to close the sleeve. A force limit of 6,000lbf was set on the linear actuator, once this limit was reached the sleeve should have fully moved. However, during this operation no sleeve movement was observed even at 6,000lbf. The anchor was gradually increased up to 20,000lbf, carefully monitoring the internal diameter measurements to ensure no deformation of the tubing. Up to 12,000lbf linear actuator force was applied downhole but only 0.3" of movement was detected. To free the sleeve from any potential debris holding it in place, the sleeve was cycled up and down slightly at 12,000lbf force. After 3 cycles of moving up and down by small increments, 13,000lbf was applied to close the sleeve and the correct distance of 3.0" was observed as seen in the below Figure. This linear actuator force was over four times greater than the ICD specification and only the use of an eLine electromechanical shifting tool would have been able to apply such high force at this depth. Confirmation of the shift was completed to verify that it was fully closed, and all measurements were validated in real-time both at the wellsite and via remote monitoring. High force was required for the complete closure of both target ICDs in this well. Due to the additional time taken to cycle the sleeves and gradually, precisely, increase to higher shifting forces the operation time extended to 8 hours for both sleeves.



Timeframe: The full operation, including multiple tool runs and verification, spanned approximately 8 hours.

Post-Job Analysis

- Real-time telemetry provided high-resolution displacement and force data.

- Each ICD shift was validated through force-displacement curves.
- The operation demonstrated tool reliability in high-deviation environments.
- The extreme force ability of the shifting tool was validated operating at over four times the expected ICD force.
- Custom ICD confirmed as suitable with operational validation due to no pre-job SIT possible.
- Final QC confirmed all ICDs were in the desired shifted state.

Well C – Coiled Tubing-Based ICD Shifting and DTS Verification

A coiled tubing intervention was performed in Well C to mechanically shift and verify the closure of two inflow control devices (ICDs) using real-time telemetry and distributed temperature sensing (DTS). The well is a high-deviation oil producer with a measured depth (MD) of 12,577 ft and a deviation of 91.87° at 10,537 ft. The operation was executed in two distinct runs: one for wellbore conditioning and the second for ICD shifting and post-shift validation. The intervention workflow is described in Fig. 3.

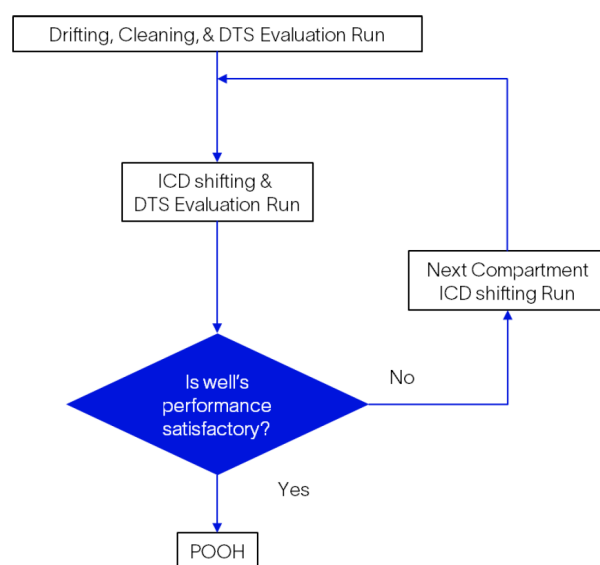


Figure 3—Engineered workflow for ICD manipulation.



Figure 4—The figure illustrates the well, which was completed with 15 ICDs and 6 oil-swellable packers across the 6-in open hole horizontal section illustrating the pre-shifting production profile. The targeted ICDs #2 & #3 were the ones contributing with most of water production.

Phase 1 – Diagnosis. The diagnostic phase involved two key runs to identify the root cause of the well's declining performance. This structured approach generated critical data to guide the intervention strategy.

The first run ensured the well was drifted and free of obstructions, preparing it for the diagnostic tools. Next, a third-party ECE-CT unit was deployed to perform production logging. This advanced system enabled real-time data capture and analysis, offering high precision in identifying problem zones.

Production logging revealed key insights into reservoir flow behavior. It identified water breakthrough from ICDs #2 and #3, which was the main driver of reduced production. Logs confirmed excessive water influx into the wellbore.

Pinpointing these problem areas was pivotal for the remediation process. Isolating the affected ICDs enabled a focused, effective intervention strategy. The rich dataset confirmed the presence and scale of the water issue, providing a clear picture of its impact on well performance.

This diagnosis emphasized the vital role of production logging in managing well integrity, especially in brownfields where flow dynamics evolve over time. By leveraging the ECE-CT unit, operators used advanced technology to collect actionable insights, allowing precise interventions and reducing operational risks.

Phase 1 demonstrated the value of detailed diagnostics paired with modern logging tools. The data gathered shaped the Phase 2 intervention, ensuring corrective actions were both targeted and effective.

Phase 2 – Intervention. This critical phase included two runs, each with specific operational goals. In the first run, the well was drifted, and the Inflow Control Devices (ICDs) were cleaned to prepare for shifting tool operations. A Fiber-Optic-Enabled Coiled Tubing (FOE-CT) unit with a high-pressure rotating nozzle was used. This setup ensured full 360-degree coverage of the tubing's internal diameter, delivering thorough cleaning.

In mature brownfields, the buildup of inorganic and organic scales is common. To address this, a dual cleanout method was applied. The high-pressure nozzle enabled hydro-mechanical scale removal, while chemical treatment dissolved mixed deposits. This combination ensured effective and complete cleaning ahead of further intervention.

Run 1 – Wellbore Conditioning and Acid Wash

The first run utilized a jet-assisted cleanout tool integrated with ACTIVE coiled tubing to prepare the ICD zones for shifting. The bottomhole assembly (BHA) included sensors for pressure, temperature, gamma ray, and casing collar locator (CCL), enabling precise depth correlation and downhole diagnostics. After reaching total depth, a sequence of fluid treatments was applied –

- **20 bbl of organic dissolver** was spotted across ICD #2 and #3 with a 60-minute soak time.
- **20 bbl of 15% HCl** was subsequently spotted with a 30-minute soak.
- **150 bbl of fresh water** was circulated to displace the acid and clean the annulus.

DTS profiles (DTS-1 to DTS-3) were recorded during shut-in, injection, and warmback phases to establish baseline thermal conditions and assess fluid placement efficiency.

Upon completion of the cleanout run, the operation proceeds to the second run, which focuses on shifting the targeted ICDs, specifically ICDs #2 and #3. The slim shifting tool, previously described, is deployed on the FOE-CT string, enabling precise manipulation of the devices. This tool is augmented with various downhole sensors that facilitate real-time, on-the-fly evaluations of the shifting process.

Run 2 – ICD Shifting and DTS Validation

The second run involved mechanical shifting of ICD #2 and ICD #3 using a hydraulic shifting tool deployed via coiled tubing. The tool was activated by generating differential pressure through controlled pumping while reciprocating across the ICD zones.

- **ICD #3 (10,856 ft MD)** was shifted through four cycles. Differential pressures ranged from 1,169 to 1,706 psi, with pumping rates between 0.8 and 1.0 bpm. Each cycle involved precise positioning and monitoring of CT weight, pressure, and temperature.
- **ICD #2 (10,688 ft MD)** was shifted through three cycles. Differential pressures reached up to 1,830 psi, confirming successful actuation. The tool was repositioned and reactivated to ensure complete closure.

DTS profiles (DTS-4 to DTS-9) were acquired before and after shifting to confirm thermal isolation and validate ICD sealing. Each DTS run involved bullheading 150 bbl of fresh water and monitoring warmback behavior over several hours.

As shown in Figure 5, sensor readings offer key insights during the ICD shifting operation. The figure illustrates changes in downhole pressure, downhole weight, and surface weight as the CT crosses ICD #3. CCL detection confirms tool position, ensuring precise alignment with the target ICD.

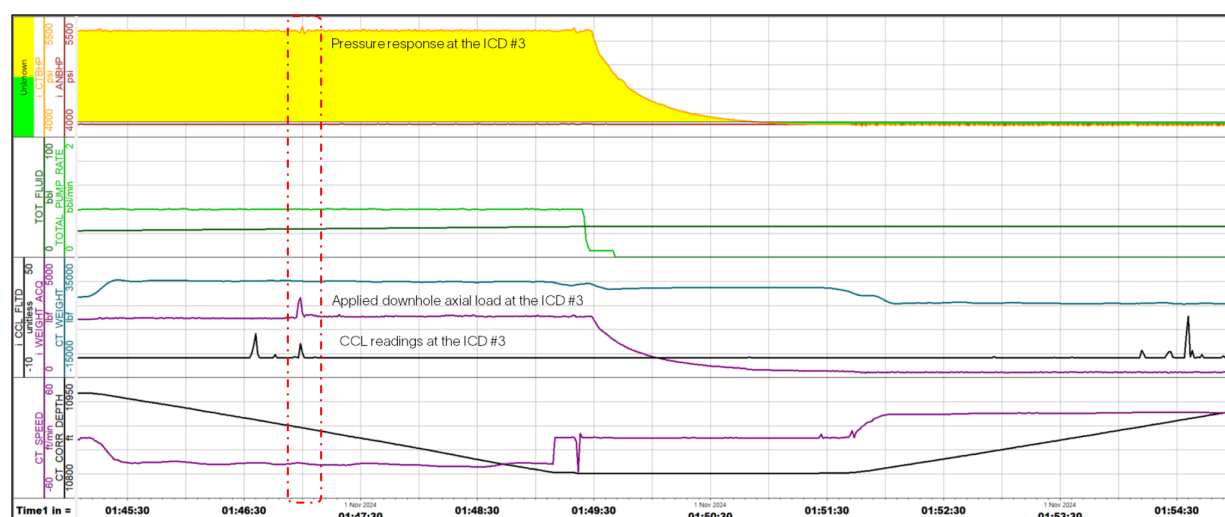


Figure 5—Several sensors indicate the proper ICD manipulation in real time, such as CCL readings, BHP, and applied downhole force. Uppermost track: in yellow is the CTBHP (psi) and in brown the Annular BHP (psi). 2nd track, in light green the pumping rate (bpm). 3rd track, in green the surface weight readings (lbf) and in purple the applied downhole force (lbf), and the CCL readings in black. The lowermost track represents in purple the CT speed (ft/min) and in black the CT depth (ft).

The integrated fiber-optic system further enhances the evaluation by capturing DTS data before and after manipulation. Pre-shifting diagnostics include two critical steps. First, during Cooldown-1, 150 barrels of surface-temperature water are pumped while monitoring the temperature profile along the CT. This step captures thermal behavior as water moves through the well.

Next, during Warmback-1, the pump is stopped to monitor temperature recovery toward geothermal levels. Zones with slower recovery absorb more water, while faster zones indicate minimal penetration or bypass.

After ICD manipulation, both tests are repeated. The Cooldown-2 log (Figure 6) shows water now reaching deeper into the horizontal lateral due to the ICD closures. The Warmback-2 log confirms new thermal patterns, revealing flow redistribution and a changed injection profile.

Evaluating ICD performance in the same run is a key advantage. By comparing pre- and post-manipulation DTS data, operators can confirm real-time effectiveness. This approach streamlines operations and helps refine production strategies in mature wells.

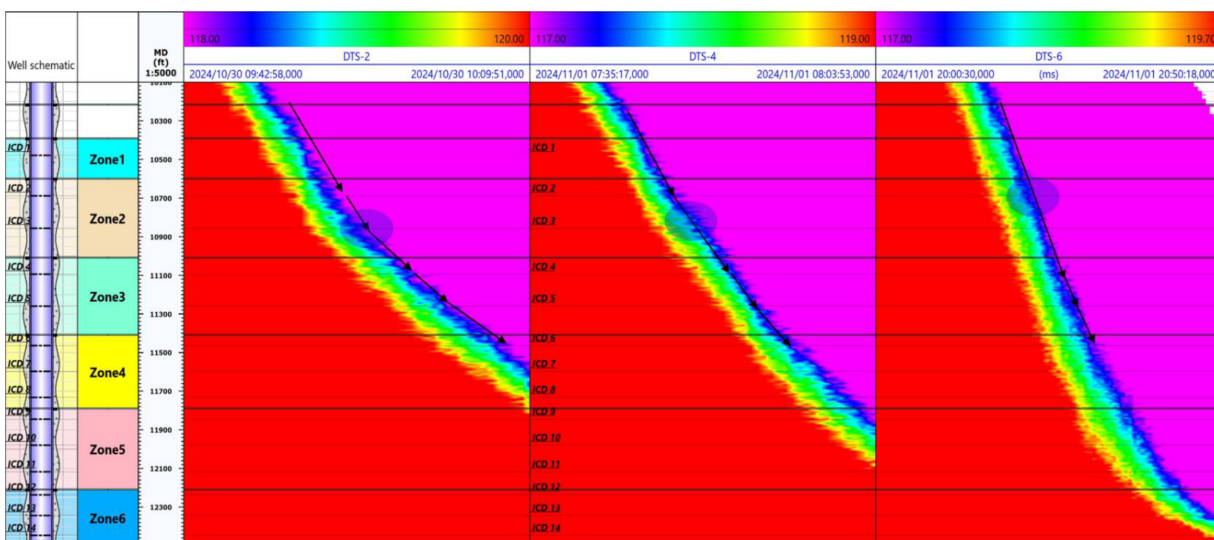


Figure 6—The pre- and post-manipulation Injection DTS profiles indicate the injected water now reaches deeper in the formation because of the restricted flow along the horizontal drain by having ICDs #2 and #3 in closed position.

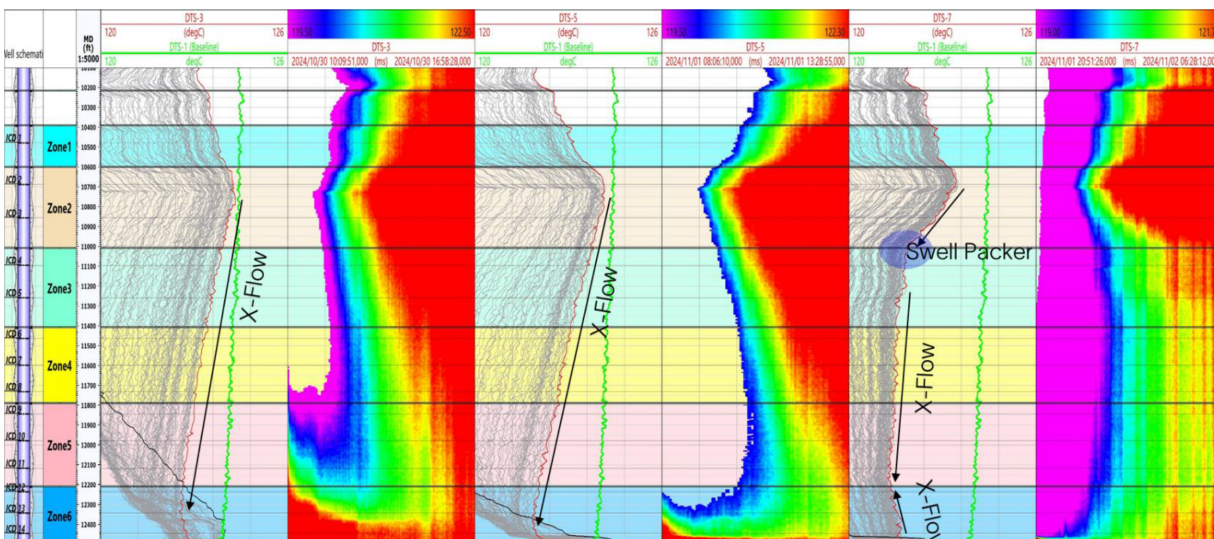


Figure 7—The pre- and post-manipulation Warmback DTS profiles indicate the crossflow profile changed along the horizontal drain by having ICDs #2 and #3 in closed position, suggesting effective closure of the ICDs and a proper seal by the swell packer.

Operational Performance and Analysis

The operation spanned from 20 October to 4 November 2024, including mobilization, rig-up, execution, and demobilization. A total of 62 operating hours were logged with zero non-productive time (NPT). Fluid volumes used included:

- 330 bbl of fresh water.
- 20 bbl of organic dissolver.
- 20 bbl of 15% HCl.
- 20 bbl of neutralizing fluid.

Real-time telemetry enabled precise control and diagnostics throughout the operation. The combination of mechanical signatures (differential pressure, CT weight) and DTS thermal profiles provided robust

confirmation of ICD closure. The intervention demonstrated the effectiveness of coiled tubing-based shifting in high-deviation wells and highlighted the value of integrated diagnostics for post-job validation.

Phase 3 – Evaluation. This phase comprises two critical steps: well testing and post-shifting production logging. These steps are essential for evaluating the success of the intervention and the effectiveness of the ICD manipulations.

Immediately after the intervention, the well was flowed to obtain a preliminary assessment of water cut reduction. During this initial testing, there was a substantial decrease in formation water production, From 47.3% to 11.5% water cut, leading to improved reservoir management and completion sustainability. The operational success resulted in a notable increase in oil production by 55% directly enhancing profitability. These results provided early confirmation of the intervention's effectiveness in optimizing reservoir performance and reducing unwanted water production.

Production Impact

Well A: Following the ICD shifting intervention, Well A exhibited a significant improvement in production performance. The water cut was reduced by approximately 18%, indicating enhanced reservoir contribution and improved fluid handling. Most notably, the oil production rate increased by 140%, reflecting successful reactivation of flow paths and optimized zonal inflow through the shifted ICDs.

Well B: Following the ICD shifting intervention, Well B revealed a substantial improvement in production performance. The water cut was reduced by approximately 8%, indicating enhanced reservoir contribution and improved fluid handling. Most notably, the oil production rate increased by 136%, reflecting successful reactivation of flow paths and enhanced zonal inflow through the shifted ICDs.

Well C: This workflow made it possible to successfully manipulate ICDs and evaluate results in real time using DTS, setting a new standard for similar operations. The team achieved ICD closure on the first attempt in an 18-month-old well with FOE-CT, confirming the reliability of the chosen tools and methods. DTS delivered high-resolution temperature data along the wellbore, providing valuable insight into ICD shifting and reservoir response. Water cut dropped from 47.3% to 11.5%, a clear sign of improved water control and lower environmental and operational risk. At the same time, oil output in well C increased by about 55%, showing the strong economic impact of the approach. The FOE-CT trial proved to be an efficient, scalable method for ICD manipulation. These results underline the technical, economic, and environmental value of the solution and its potential for wider application across the field.

Conclusions

The 2024 ICD shifting campaign across three ADNOC Onshore wells has demonstrated a transformative leap in completion optimization, enabled by the integration of real-time diagnostics, precision hydraulic actuation, and advanced conveyance technologies. The successful deployment of electromechanical tools via electric wireline and fiber-optic-enabled coiled tubing has not only resolved longstanding challenges associated with blind overpull and force transmission inefficiencies but also established a repeatable, data-driven workflow for ICD manipulation in high-deviation environments. The interventions yielded measurable production improvements in selected wells and provided conclusive verification of ICD status through downhole telemetry and DTS. These results affirm the operational reliability, scalability, and reservoir impact of the approach, positioning it as a benchmark for future inflow control strategies in mature fields. The methodology offers a robust framework for unlocking bypassed reserves, reducing water production, and extending asset life setting the stage for broader adoption across similar legacy completions worldwide.

These case studies highlight the importance of integrating advanced diagnostic tools and innovative technologies to address complex reservoir challenges. By combining meticulous planning with state-of-the-art systems like electromechanical ICD shifting tools, e-line tractors, DTS and FOE-CT, the project

team achieved a comprehensive evaluation and successful remediation of the well's issues. The outcomes not only improved immediate production metrics but also provided a scalable solution for managing flow dynamics in mature fields.

The success of this workflow demonstrates its potential as a best practice for well interventions, offering a sustainable and economically sound approach to enhancing reservoir performance.

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